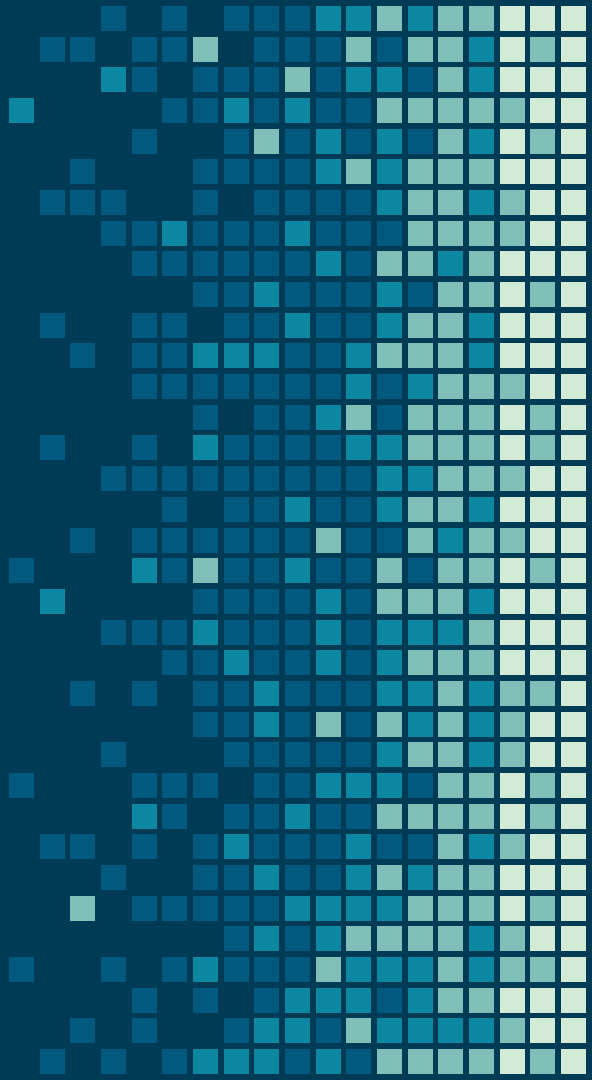
EMBEDDED CONTROL

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Projects assignment

- Projects assigned to groups at around mid of the course
- Two possibilities:
  - Use one of the plants available at MOBIREC lab
  - Projects can also be developed without using the lab plants (subject to approval)

In case you will use plants @ MOBIREC:

- **No equipment leaves the lab**, no exceptions made
- Access to the MOBIREC lab granted according to a calendar
- Groups using lab plants **must demonstrate steady progress**
- For MOBIREC projects: **deadline 31 October**

Projects **mandatory** checkpoints

- All groups working on projects will be monitored
- Checkpoints are intended to ensure steady progress and reduce integration risks
- Short online checkpoints will be scheduled bi-weekly
- Once the project is officially started the checkpoints are **MANDATORY**
- The meetings will monitor:

1. if the group is working on the project
2. if there are blocking issues to be solved

- Groups failing to show up at checkpoints twice (1 month period) without notice are considered not interested to carry out the project with provided plants
- Those plants will be reassigned to other groups

Grading

- Both theoretical understanding and practical implementation are essential to pass the course
- **Project grade (PG)**: 0 to 10 (min 4) - Assessed through project deliverable + 1 question at the oral exam
- **Theory grade (TG)**: 0 to 20 (min 10)- Assessed through the oral exam

Grade = PG + TG

Project deliverables

- The code to be provided in a github repository
- A short report in PDF (we will provide a template)
- Optional (but recommended): videos showing the plant in operation. Those can be links in the report.

“Available plants @ MOBIREC

Project bare minimum requirements

- Have a stable closed loop system
- Use interrupts at least for one peripheral
- Use UART over USB to communicate with the plant:
  - Change controller parameters
  - Send setpoints
  - Send commands (start/stop)

Inverted pendulum

- **Bare minimum objective**: stabilize the pendulum in the upright position
- **Plus**: swing-up from the downright position
- **Actuator**: brushless motor + ESC
- **Sensor**: rotary relative encoder
- **Power**: through bench power supply
- **To be done**: wiring your nucleo board to the system
- **Big plus**: design and implement a shield for the nucleo with connectors



Photo representative of the real system

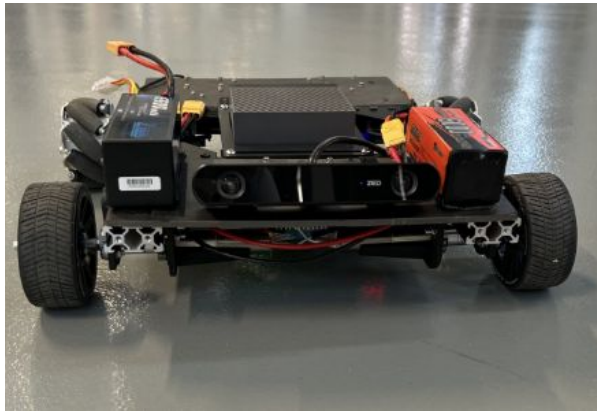
Extruder control

- **Bare minimum objective**: stabilize the temperature to a given setpoint. Implement the relevant Gcode commands for commanding the extruder (temperature, extrusion) sent through UART
- **Actuators**: Nema + driver, heating cartridge + mosfet
- **Sensor**: thermistor
- **Power**: bench power supply
- **To be done**: wiring your nucleo board to the system
- **Big plus**: design and implement a shield for the nucleo with connectors



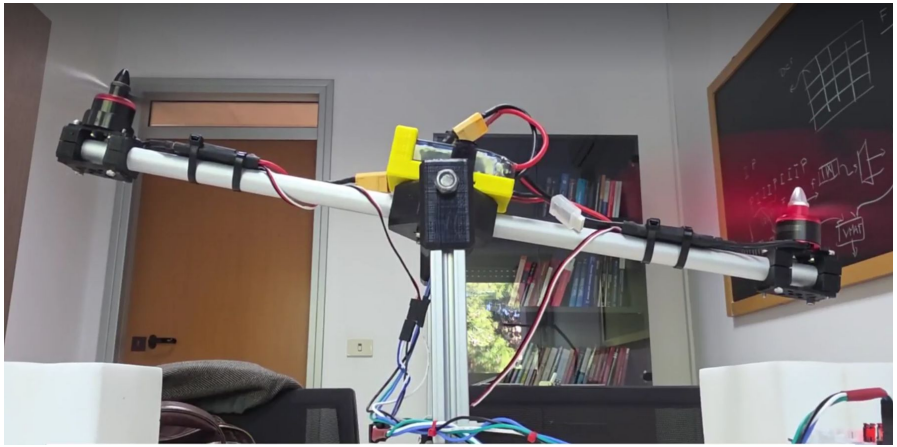
Traction Control Pluto

- **Bare minimum objective**: implement traction control to avoid slippage regardless of the setpoint profiles provided
- **Actuators**: DC motors + drivers
- **Sensors**: CUI rotary encoders, IMU, shunt resistor
- **Power**: batteries



1 DoF Drone

- **Bare minimum objective**: stabilization to a given angle
- **Plus**: comparison between different control algorithms, automatic tuning, trajectory control
- **Actuators**: 2x brushless motors + ESC
- **Sensor**: onboard IMU
- **Power**: batteries



STEVAL Drone

- **Objective**:
  attitude control
- **Actuators**: 4x dc motors + drivers
- **Sensor**: onboard IMU + barometer
- **Power**: batteries



Levitating ping pong ball

- **Bare minimum objective**: ball height regulation, trajectory control (f.i., sinusoid reference)
- **Actuator**: 1 dc motor + driver
- **Sensor**: ToF sensor (VL53L0X) - I2C
- **Power**: bench power supply
- **To be done**: wiring your nucleo board to the system
- **Big plus**: design and implement a shield for the nucleo with connectors



Jupyter robot

- **Bare minimum objective**: stabilize the robot in the upright position (don't break it!)
- **Plus**: compare different controllers, implement velocity control
- **Actuator**: 2x Nema 23 motors + driver
- **Sensors**: IMU, 2x CUI relative rotary encoders
- **Power**: battery

